

## IP Lab

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Windows PowerShell
Tracing route to gaia.cs.umass.edu [128.119.245.12]
over a maximum of 30 hops:

  1  5 ms  7 ms  5 ms  10.20.0.1
  2  5 ms  4 ms  4 ms  192.168.46.1
  3  13 ms 10 ms 46 ms 192.168.45.9
  4  18 ms 76 ms 6 ms 205.189.187.3
  5  7 ms  9 ms 10 ms 66-207-208-113.beanfield.net [66.207.208.113]
  6  *  * 3970 ms Et8-1.lsr01.494ChampagneDr01.YYZ.beanfield.com [76.9.218.228]
  7  9 ms  * 10 ms et33-1.lsr01.3389SteelesAvE01.yyz.beanfield.com [207.66.123.175]
  8  73 ms 6 ms 22 ms be320.dr01.3389steelesave01.yyz.beanfield.com [199.167.154.210]
  9  17 ms 12 ms 11 ms po320.lsr01.3389steelesave01.yyz.beanfield.com [199.167.154.211]
 10  *  *  * Request timed out.
 11  *  *  * Request timed out.
 12 378 ms 250 ms 32 ms 38.104.158.174
 13 138 ms 140 ms 117 ms hu0-6-0-5.ccr32.yyz02.atlas.cogentco.com [38.104.158.173]
 14 13 ms 15 ms 19 ms be3260.ccr22.ymq01.atlas.cogentco.com [154.54.42.90]
 15 29 ms 30 ms 26 ms be2104.ccr22.alb02.atlas.cogentco.com [154.54.43.22]
 16 23 ms 30 ms 21 ms be8028.rcr71.orh02.atlas.cogentco.com [154.54.169.217]
 17 74 ms  * 81 ms be2191.rcr71.orh01.atlas.cogentco.com [154.54.160.70]
 18 25 ms  * 44 ms 38.32.33.83
 19 547 ms 175 ms  * 69.16.0.9
 20 29 ms 37 ms 36 ms 69.16.3.0
 21 44 ms 29 ms 31 ms core2-rt-et-4-3-0.gw.umass.edu [192.80.83.105]
 22 36 ms 34 ms 51 ms n5-rt-1-1-et-10-0-0.gw.umass.edu [128.119.0.10]
 23 104 ms 51 ms 49 ms 128.119.77.235
 24  *  *  * Request timed out.
 25 33 ms 30 ms 29 ms cics-rt-xe-0-0-0.gw.umass.edu [128.119.3.32]
 26  *  *  * Request timed out.
 27 31 ms 109 ms 32 ms gaia.cs.umass.edu [128.119.245.12]

Trace complete.

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*Wi-Fi
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icmp

No. Time Source Destination Protocol Length Info
497 1.448005700 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=5/1280, ttl=1 (no response ...)
498 1.453614600 10.20.0.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
499 1.454612400 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=6/1536, ttl=1 (no response ...)
500 1.462308900 10.20.0.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
501 1.464228700 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=7/1792, ttl=1 (no response ...)
502 1.469521700 10.20.0.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2476 7.418541500 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=8/2048, ttl=2 (no response ...)
2479 7.424409100 192.168.46.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2480 7.425577200 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=9/2304, ttl=2 (no response ...)
2484 7.429702000 192.168.46.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2485 7.430920300 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=10/2560, ttl=2 (no response ...)
2487 7.435308900 192.168.46.1 10.20.11.119 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2534 7.486462900 192.168.46.1 10.20.11.119 ICMP 70 Destination unreachable (Communication administratively filtere...)
2969 8.988475600 192.168.46.1 10.20.11.119 ICMP 70 Destination unreachable (Communication administratively filtere...)
3650 10.517666200 192.168.46.1 10.20.11.119 ICMP 70 Destination unreachable (Communication administratively filtere...)
4667 12.986453900 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=11/2816, ttl=3 (no response ...)
4674 13.000012400 192.168.45.9 10.20.11.119 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
4676 13.001101100 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=12/3072, ttl=3 (no response ...)
4677 13.011840600 192.168.45.9 10.20.11.119 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
4678 13.012982500 10.20.11.119 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=13/3328, ttl=3 (no response ...)

Frame 497: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on 0
Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Cisco_2e:c3:
Internet Protocol Version 4, Src: 10.20.11.119, Dst: 128.119.245.12
Internet Control Message Protocol
0000 9c 09 8b 2e c3 d3 48 45 e6 8b f6 df 08 00 45 00 ... HE ..... E
0010 00 5c 03 3f 00 00 01 01 00 00 0a 14 0b 77 80 77 ... ? ..... w w
0020 f5 0c 08 00 f7 f9 00 01 00 05 00 00 00 00 00 00 ... .....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ... .....
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ... .....
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ... .....
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ... .....

wireshark -Wi-Fi99N6Q3.pcapng
Packets: 64095 · Displayed: 152 (0.2%) · Dropped: 0 (0.0%) · Profile: Default

```

1. My computers IP address: 10.20.11.119
2. Upper layer protocol field value = ICMP (1) = decimal value 1
3. Header Length = 20 bytes  
Total Length = 92 bytes  
Payload = 92 - 20 = 72 bytes

## 4. No, because:

Flags: 0x0 → More Fragments bit = NOT set

Fragment Offset: 0 → starts at beginning, no offset

Thus, Not fragmented because the More Fragments flag is 0 and Fragment Offset is 0

The screenshot shows a Wireshark capture of ICMP Echo (ping) requests. The packet list displays several requests from source 10.20.11.119 to destination 128.119.245.12. The packet details for packet 4678 (ICMP Echo request) are expanded, showing the following fields:

- Version: 4
- Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 92
- Identification: 0x0340 (832)
- Flags: 0x0
- Fragment Offset: 0
- Time to Live: 1
- Protocol: ICMP (1)

The packet bytes pane shows the raw data of the ICMP Echo request, including the ID field (0x033f) and the TTL field (0x01).

## 5. Identification: 831 → 832 (increments by 1)

That's the main one changing here. TTL will change too once traceroute moves to the next hop.

## 6. Must stay constant:

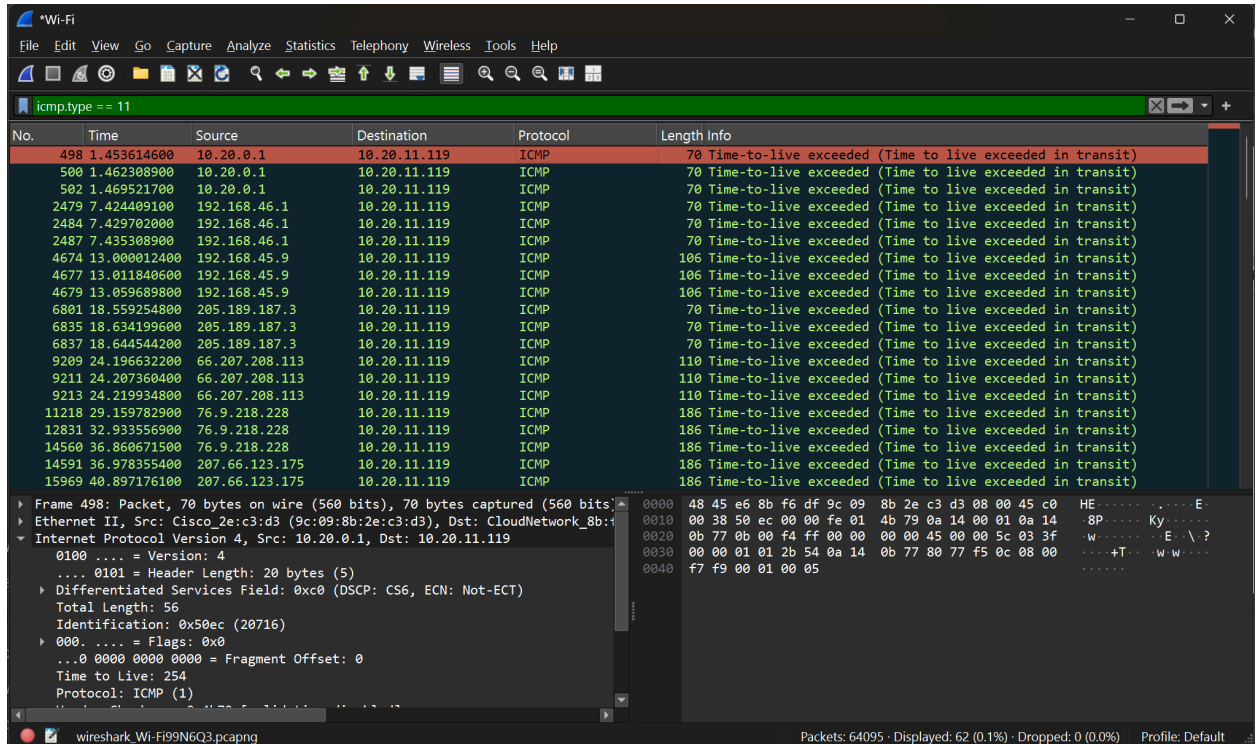
- Source/Destination IP (same machine, same target)
- Protocol (always ICMP)
- Header Length (no options added)

Must change:

- TTL (core mechanic of traceroute)
- Identification (each datagram gets unique ID)

## 7. 0x033f (831) → 0x0340 (832)

The Identification field increments by 1 with each new datagram sent



8. Identification: 0x7081 (28801)

TTL: 38

9. No, because:

- The Identification is the destination server's own counter, completely independent from yours
- TTL of 38 means the server started with a standard value (probably 64) and lost 26 hops getting to you

ip-ethereal-trace-1.pcap

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ip.flags.mf == 1 || ip.frag\_offset > 0

| No. | Time      | Source         | Destination   | Protocol | Length | Info                                                               |
|-----|-----------|----------------|---------------|----------|--------|--------------------------------------------------------------------|
| 92  | 28.441511 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32f9) [Reassemb... |
| 93  | 28.442185 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=30467/887, ttl=1 (no respon...  |
| 94  | 28.462264 | 10.216.228.1   | 192.168.1.102 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)           |
| 95  | 28.470668 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32fa) [Reassemb... |
| 96  | 28.471338 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=30723/888, ttl=2 (no respon...  |
| 97  | 28.490663 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32fb) [Reassemb... |
| 98  | 28.491323 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=30979/889, ttl=3 (no respon...  |
| 99  | 28.520729 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32fc) [Reassemb... |
| 100 | 28.521393 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=31235/890, ttl=4 (no respon...  |
| 101 | 28.530213 | 24.218.0.153   | 192.168.1.102 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)           |
| 102 | 28.540758 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32fd) [Reassemb... |
| 103 | 28.541476 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=31491/891, ttl=5 (no respon...  |
| 104 | 28.570848 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32fe) [Reassemb... |
| 105 | 28.571603 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=31747/892, ttl=6 (no respon...  |
| 106 | 28.590801 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=32ff) [Reassemb... |
| 107 | 28.591465 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=32003/893, ttl=7 (no respon...  |
| 108 | 28.597502 | 24.128.190.197 | 192.168.1.102 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)           |
| 109 | 28.620895 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=3300) [Reassemb... |
| 110 | 28.621558 | 192.168.1.102  | 128.59.23.100 | ICMP     | 562    | Echo (ping) request id=0x0300, seq=32259/894, ttl=8 (no respon...  |
| 111 | 28.640865 | 192.168.1.102  | 128.59.23.100 | IPv4     | 1514   | Fragmented IP protocol (proto=ICMP 1, off=0, ID=3301) [Reassemb... |

Frame 102: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface 0

Ethernet II, Src: ActiontecEle\_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysGr\_00:0c:29:1a:00:00

Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.59.23.100

0100 .... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 1500

Identification: 0x32fd (13053)

001. .... = Flags: 0x1, More Fragments

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 5

Protocol: ICMP (1)

Header length in 32-bit words (ip.hdr.len), 4 bits

Packets: 380 - Displayed: 284 (74.7%)

Profile: Default

10. Yes! It was split into 2 fragments, packets #92 and #93. We can see at the bottom of #93 it confirms: 2 IPv4 Fragments (2008 bytes): #92(1480), #93(528)

11. How we know it's fragmented: MF flag = 1 meaning more pieces follow

How we know it's the FIRST fragment: Fragment Offset = 0

Total Length = 1500 bytes

12. How we know it's NOT the first fragment: Fragment Offset = 1480 (not zero)

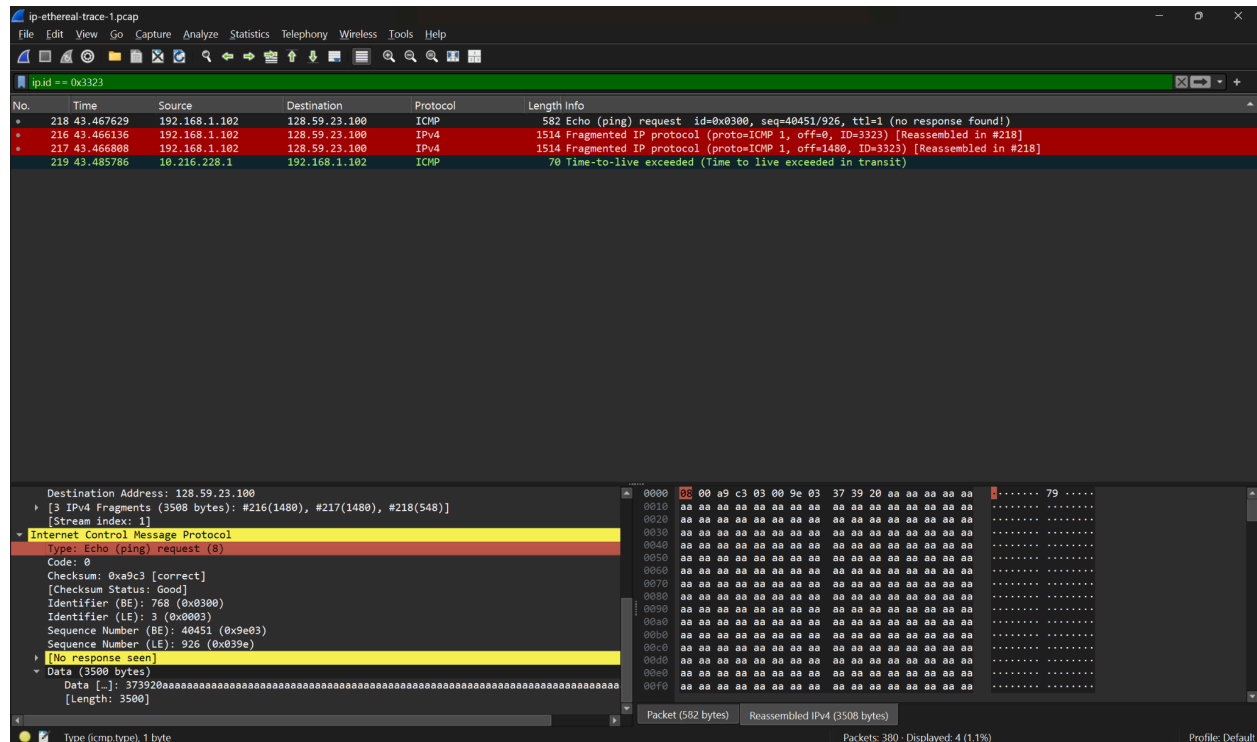
Are there more fragments? No, MF flag = 0 (no more fragments)

Total Length = 548 bytes

| Field           | #92    | #93    |
|-----------------|--------|--------|
| Flags (MF)      | 1      | 0      |
| Fragment Offset | 0      | 1480   |
| Total Length    | 1500   | 548    |
| Header Checksum | 0x077b | 0x2a7a |

Everything else (Identification, Source, Destination, Protocol) stayed the same.

IP Lab



14. 3 fragments, packets #216, #217, and #218, totalling 3508 bytes.

### 15. Fields that change in the IP header among the fragments

|                 |            |            |            |         |
|-----------------|------------|------------|------------|---------|
| Field           | #216 (1st) | #217 (2nd) | #218 (3rd) |         |
| Fragment Offset | 0          | 1480       | 2960       |         |
| MF Flag         | 1          | 1          | 0          |         |
| Total Length    | 1500       | 1500       | 548        |         |
| Header Checksum | Changes    |            | Changes    | Changes |

Fields that stay the same:

- Identification: 0x3323 (same across all 3 fragments — this is how they are reassembled)
- Source/Destination IP: Same
- Protocol: Same